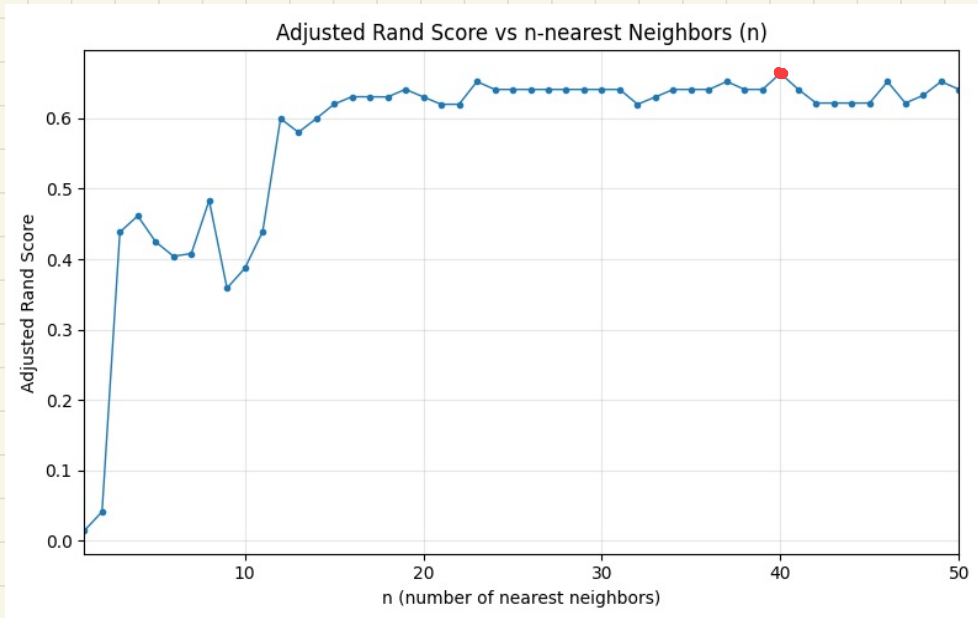


“cluster - gr”

iris - knn

```
=== Spectral Clustering 結果 (最佳 k) ===  
best k: 40  
Adjacency matrix edge count: 3659  
Adjusted Rand Score (與真實標籤比較): 0.6639  
Silhouette Score: 0.4535  
集群 0: 47 個樣本  
集群 1: 53 個樣本  
集群 2: 50 個樣本
```



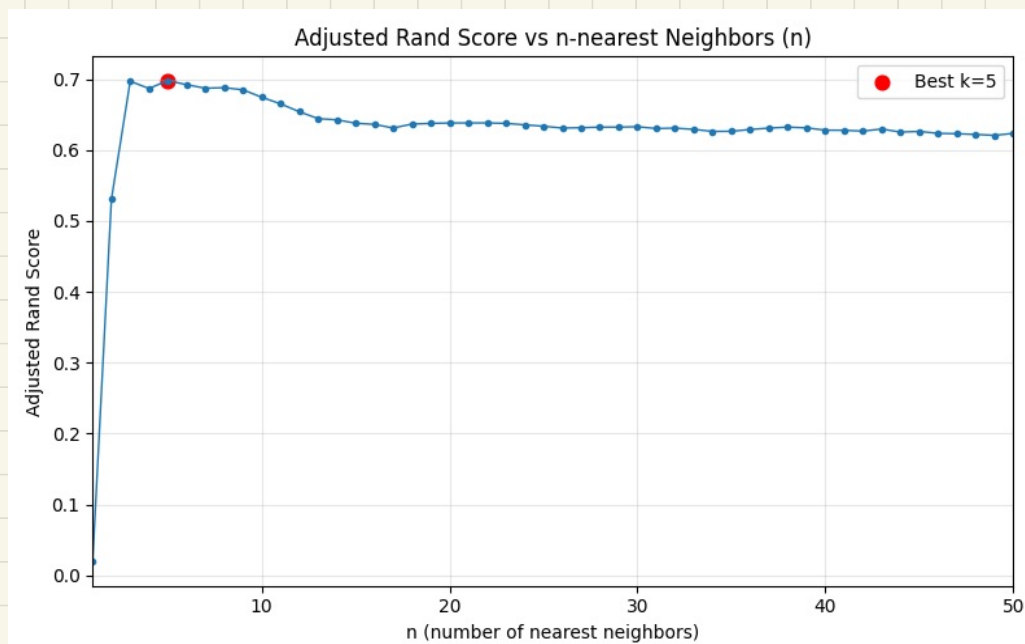
iris - knn - kernel

```
0.00016030 -> 0.6639  
0.00028505 -> 0.6639  
0.00050691 -> 0.6639  
0.00090142 -> 0.6639  
0.00160298 -> 0.6639  
0.00285055 -> 0.6639  
0.00506907 -> 0.6639  
0.00901422 -> 0.6639  
0.01602979 -> 0.6639  
0.02850545 -> 0.6639  
0.05069066 -> 0.6639  
0.09014216 -> 0.6639  
0.16029795 -> 0.6639  
0.28505454 -> 0.6410  
0.50690661 -> 0.6410
```

```
=== Spectral Clustering 結果 (最佳 k) ===  
best k: 40  
Adjacency matrix edge count: 3659  
Adjusted Rand Score (與真實標籤比較): 0.6639  
Silhouette Score: 0.4535  
集群 0: 47 個樣本  
集群 1: 53 個樣本  
集群 2: 50 個樣本
```

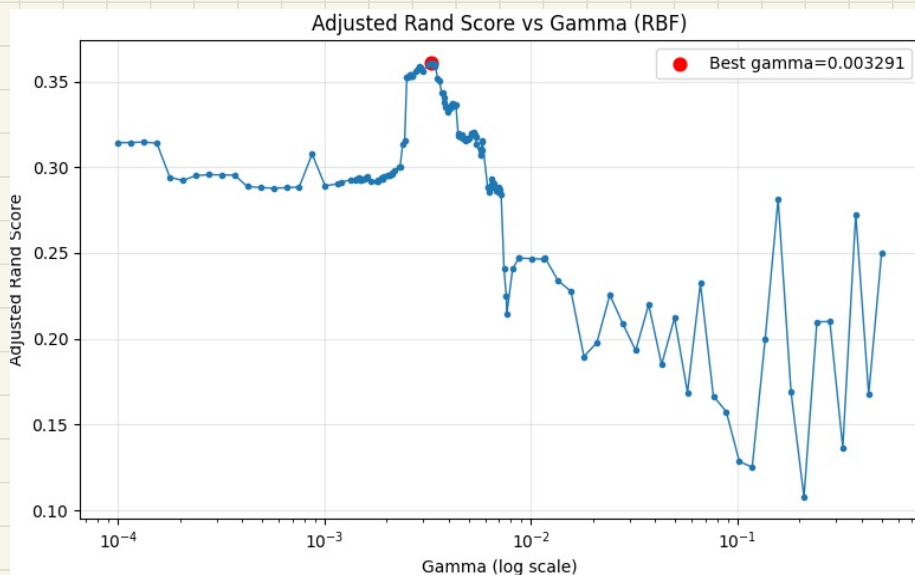
```
=== Masked RBF + Spectral Clustering 結果 ===  
base gamma: 0.160298  
best gamma: 0.016030  
Adjusted Rand Score (與真實標籤比較): 0.6639  
Silhouette Score: 0.4535  
Masked similarity edge count: 3659
```

digit - knn - kernel (25 維)



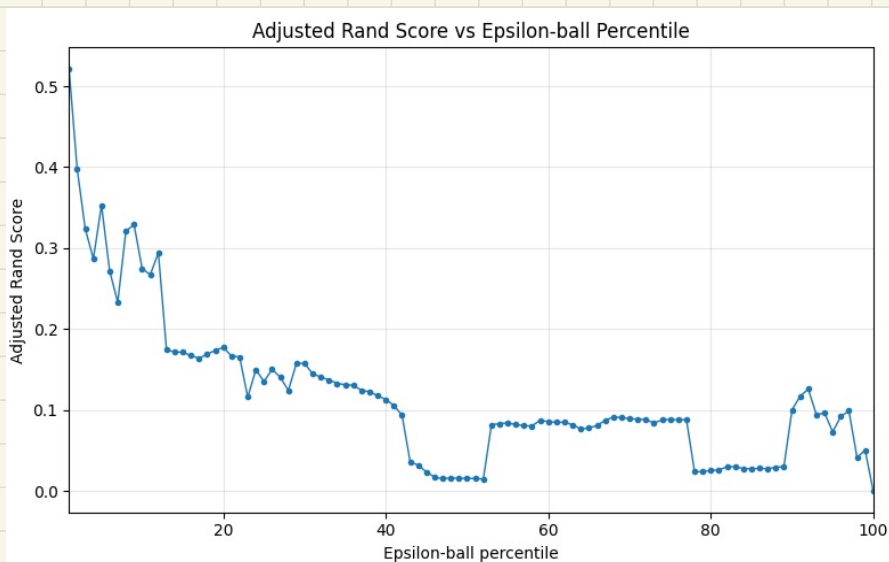
```
base gamma: 0.012323
gamma -> ARI
0.00001232 -> 0.6978
0.00002191 -> 0.6978
0.00003897 -> 0.6978 ← original
0.00006930 -> 0.6978
0.00012323 -> 0.6978
0.00021914 -> 0.6978
0.00038970 -> 0.6978
0.00069299 -> 0.6980 ← improve
0.00123234 -> 0.6951
0.00219144 -> 0.6948
0.00389699 -> 0.6957
0.00692993 -> 0.6960
0.01232335 -> 0.6960
best gamma: 0.000693
Adjusted Rand Score (與真實標籤比較): 0.6980
```

digit - kernel (25 维)



(很惨)

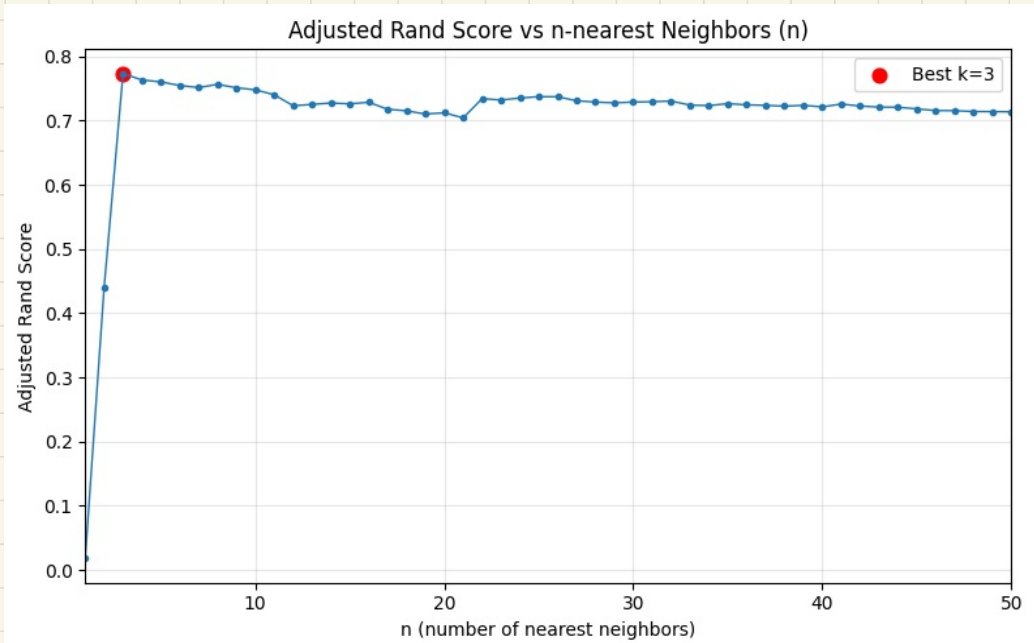
降维效果(?)



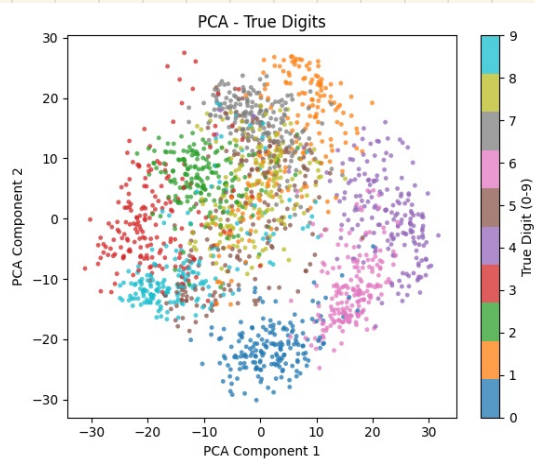
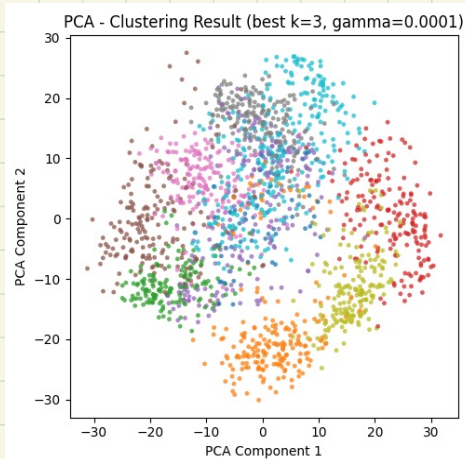
iris - kmean

```
=== KMeans 結果 ===  
Adjusted Rand Score : 0.6201  
Silhouette Score   : 0.4599  
集群 0: 53 個樣本  
集群 1: 50 個樣本  
集群 2: 47 個樣本  
PS C:\Users\danie\Desktop\project> py .\Grok_code.py  
=== KMeans 結果 ===  
Adjusted Rand Score : 0.7302  
Silhouette Score   : 0.5528  
集群 0: 62 個樣本  
集群 1: 50 個樣本  
集群 2: 38 個樣本
```

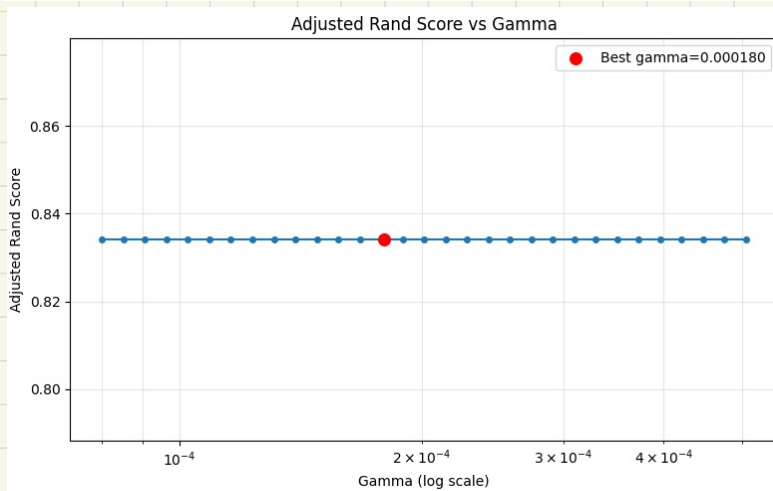
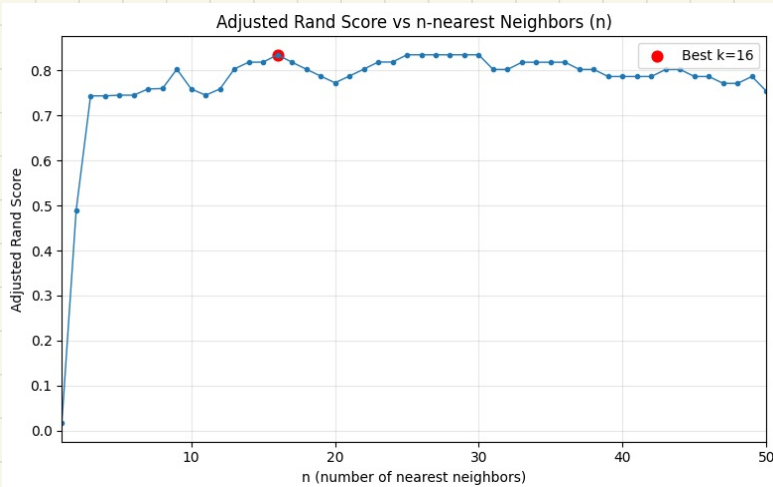
digit - knn - kernel



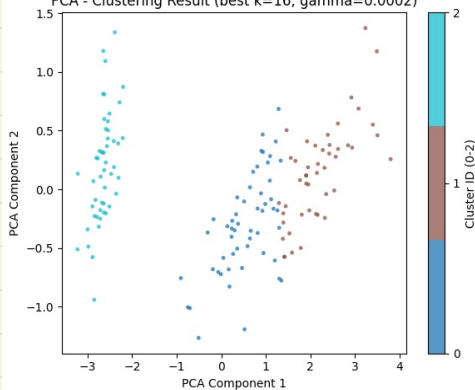
```
0.00000100 -> 0.8306
0.00000141 -> 0.7733
0.00000250 -> 0.7733
0.00000445 -> 0.7733
0.00000791 -> 0.7733
0.00001407 -> 0.7733
0.00002502 -> 0.7733
0.00004450 -> 0.7733
0.00007913 -> 0.8306
0.00014072 -> 0.8327
0.00025023 -> 0.7805
0.00044498 -> 0.8326
0.00079130 -> 0.8233
0.00140716 -> 0.8211
0.00250232 -> 0.7162
best gamma: 0.000141
Adjusted Rand Score (與真實標籤比較): 0.8327
```



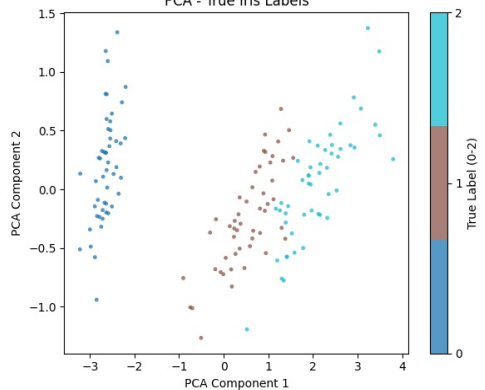
iris - knn - kernel



PCA - Clustering Result (best k=16, gamma=0.0002)



PCA - True Iris Labels



Question

(1) 降維效果？

(2) gamma 值

1. α 當基準不好

2. 如何搜尋